Homework

Solve the following systems of equations using the Gaussian elimination method with back substitution.

In each case, answer these questions

(i) What is the solution?
(ii) What are the corresponding lower and upper triangular matrices L and U?
(iii) Is A = LU?
(iv) Find the determinant of A.

1. $\mathrm{Ax} = \begin{bmatrix} 4 & 2 & 3 \\ 2 & -4 & -1 \\ -1 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix}$

2. Ax = $\begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & -1 \\ -3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$

3. Ax = $\begin{bmatrix} 4 & 2 & 3 \\ 2 & -4 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \\ -2 \end{bmatrix}$